# Vietnamese – German University

## Faculty of Engineering

## Computer Science Program

# PRACTICAL TRAINING REPORT

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|---|---|
| Company: | AIVISION JOINT STOCK COMPANY |
| Intern period: | 11/03/2026 – 10/07/2026 |
| Supervisor: | Dr. Vo Truong Giang |


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| Intake: | 2022 |


2026, Viet Nam

# Declaration

I hereby declare that this internship report has been created solely by myself. I affirm that I have not utilized any sources other than those specified in the references section of this report.

Furthermore, I assert that all information presented within this report is based on my own observations, experiences, and analyses during the course of my internship at AIVISION JOINT STOCK COMPANY. Any contributions or assistance provided by others have been acknowledged in the appropriate section of this report.

I understand the importance of academic integrity and take full responsibility for the originality and authenticity of this work.

| **Author** | **Company Representative** |
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| ______________________ | ______________________ |
| Full Name: ______________ | Full Name: ______________ |
| Date: __________________ | Date: __________________ |

# Table of Contents

# Abstract

This report documents a four-month practical training phase as a Business Analyst Intern at AIVISION Joint Stock Company, an AI and software development firm based in Vietnam that builds and outsources intelligent applications for major technology clients. The internship took place from March 11 to July 10, 2026.

The core challenge addressed throughout the internship was bridging the communication gap between external customers and the internal development team. Operating in the role of a Business Analyst Intern, the assigned tasks spanned most of the software development lifecycle, including requirements gathering, wireframe and UI designing, project management, documentation, and quality control.

The work was carried out using an industry-standard toolset comprising Azure DevOps for project tracking and collaboration, Figma and FigJam for wireframing and mockup creation, and React with TypeScript for the detailed implementation of front-end designs. Microsoft Office and other industry-standard AI tools were also utilized for Business Analysis tasks, such as NotebookLM for generating meeting minutes and Claude/ChatGPT for daily task management and documentation.

As a result of this internship, end-to-end solutions were successfully delivered by translating customer requirements into clear, actionable specifications for the development team. Through this experience, practical competencies were developed in requirements gathering, project management, and wireframe design — skills that are essential to the role of a Business Analyst in a modern software organization.

# Chapter 1

# Introduction

Practical training is a compulsory component of the Computer Science program at the Vietnamese-German University (VGU), designed to provide students with first-hand exposure to professional working environments before graduation. This report documents my four-month internship placement undertaken at AIVISION Joint Stock Company from March 11 to July 10, 2026.

I came across this opportunity through a Business Analyst Intern position advertised on LinkedIn. With no prior professional experience in the role, I was drawn to the position for several practical reasons: it was a manageable four-month commitment, the office was conveniently close to home, the internship came with a reasonable allowance, and the company had a strong professional reputation. After submitting my CV, I was invited for an interview and was pleased to receive an offer shortly after.

My internship at AIVision began with an onboarding period where I assisted a senior Business Analyst on smaller-scale outsourcing software projects for clients such as GeneSolutions. This initial phase helped me build foundational skills in requirements gathering, wireframing, and documentation in a supported environment. After roughly one month, I took on greater independence and began managing three to five larger AI-driven projects simultaneously for major enterprise clients, including Masan and INNOVA US.

Going into this placement, my main personal goal was straightforward: to learn what it actually means to work as a Business Analyst — not from textbooks, but through real interactions with clients and development teams, solving real problems and delivering real solutions.

The remainder of this report is structured as follows. Chapter 2 provides a profile of AIVision as an organization. Chapter 3 outlines the key problems and challenges that I encountered during the placement. Chapter 4 describes the technologies and tools exercised throughout my internship. Chapter 5 details the tasks assigned and the methodology applied to complete them. Chapter 6 reflects on intra-organization communication and team dynamics. Chapter 7 presents the results produced and the skills developed. Finally, Chapters 8 and 9 offer my personal reflections and concluding remarks, respectively.

# Chapter 2

# Profile of the Organization

## 2.1 Company Overview and History

AIVision Joint Stock Company (aivision.vn) is a leading artificial intelligence and digital transformation enterprise headquartered in Ho Chi Minh City, Vietnam. The company was founded in January 2023 by Dr. Giang Vo, an AI scientist with fifteen years of international research and industry experience in Singapore and the United States. Established with the vision of bringing world-class AI technology to the Vietnamese market, AIVision grew at a remarkable pace, evolving into a million-dollar startup within its first year of operation.

## 2.2 Leadership and Workforce

AIVision maintains a highly specialised technical workforce of over 150 AI experts. The leadership and research teams are composed of elite professionals, including scientists holding PhD degrees from the National University of Singapore (NUS) and the University of Cambridge, as well as senior engineers with prior experience at major global technology firms such as NVIDIA (USA), alongside specialists from Germany and Singapore. This concentration of international talent underpins the company's capacity to develop proprietary, research-grade AI solutions at a commercially competitive pace.

## 2.3 Core Products and Services

AIVision develops proprietary AI models, retaining 100% intellectual property rights, optimised for the Southeast Asian market through the use of Edge Computing and real-time data synchronisation. The company offers four core enterprise solution lines:

- **AI Display Scoring (AIVision Display Tracking).** An automated Computer Vision system that evaluates retail product displays by recognising products, counting Stock Keeping Units (SKUs), and scoring Planogram compliance. The system incorporates 360-degree panorama stitching and fraud detection, achieving 97–100% accuracy with a maximum processing latency of under one second.

- **Agentic AI.** A next-generation multi-agent platform integrating Large Language Models (LLMs) such as GPT-4 and Llama. Acting as a "Digital Operations Copilot", it automates complex business workflows, customer service interactions, document summarisation, and data analysis through autonomous planning and tool use.
- **AI Camera Analytics.** A smart video analysis system integrated into existing surveillance infrastructure, designed to automatically detect retail fraud, monitor factory security, perform Automatic Number Plate Recognition (ANPR), and generate foot-traffic heatmaps.
- **Custom AI Software.** Tailored enterprise platforms including an AI HR Platform that automates 90% of the recruitment process via CV parsing and intelligent candidate matching, as well as Computer Vision applications for the healthcare sector to track and analyse dermatological treatments.

## 2.4 Key Achievements and Client Portfolio

AIVision's solutions have been validated in large-scale, real-world production environments. The company serves a portfolio of billion-dollar corporations and international brands, including Masan, Pepsi, Marico, VNG, and INNOVA US. Most notably, AIVision successfully deployed its AI Display Scoring platform across Masan's network of over 500,000 retail points of sale, building a system capable of handling more than 5,000 concurrent users while maintaining a 99% customer satisfaction rate.

## 2.5 Global Presence and Expansion Strategy

While headquartered in Ho Chi Minh City, AIVision maintains a strong international presence with physical offices in Singapore, Bangkok (Thailand), and Texas (USA). The organization is actively executing a global expansion strategy with the ambition of becoming a strategic AI partner in markets such as Australia, Mexico, and the broader United States by 2029.

# Chapter 3

# Problems Identified and Challenges Faced

This chapter outlines the core business problems that motivated the projects undertaken during the internship, as well as the practical challenges encountered while executing the role of a Business Analyst.

## 3.1 Business Problems Addressed

### 3.1.1 AI Display Tracking (Masan, Syngenta, P&G)

For large Fast-Moving Consumer Goods (FMCG) companies operating across massive retail networks, manually tracking product displays across thousands of points of sale is practically infeasible. Clients such as Masan, Syngenta, and P&G engaged AIVision's AI Display Scoring platform to address several interconnected problems. First, manually identifying products and counting SKUs on store shelves was slow, error-prone, and could not scale. Second, ensuring that retail outlets actually arranged products according to the brand's agreed Planogram layout was difficult to enforce and verify at scale. Third, detecting display fraud — such as misreported compliance — across a vast network was nearly impossible without an automated system. Finally, gathering intelligence on competitor product placements required human fieldwork that was both costly and inconsistent. AIVision's Computer Vision platform, incorporating 360-degree panorama stitching and automated fraud detection, was the technical solution proposed to address all of these problems simultaneously.

### 3.1.2 Agentic AI (INNOVA US)

INNOVA US is an American automotive technology company facing two core data challenges in the automotive diagnostics domain. First, valuable technical knowledge — spread across thousands of unstructured manufacturer documents, OEM websites, and automotive forums — could not be efficiently extracted, normalised, and made queryable at scale. Second, there was no automated system to collect and categorise the vast volume of owner-reported Q&A discussions scattered across the internet for specific vehicle models. Both problems required AI-driven pipelines to transform unstructured, dispersed data into structured, actionable datasets suitable for powering diagnostic tools.

### 3.1.3 Software Development (GeneSolutions)

GeneSolutions, a Vietnamese healthcare company specialising in genetic testing and prenatal care, required a comprehensive mobile application to support expectant mothers throughout their pregnancy journey and beyond. The underlying problems were numerous: pregnant women lacked access to timely, personalised medical guidance outside clinic hours; parents had difficulty tracking critical prenatal milestones and newborn vaccination schedules; financial planning for pregnancy costs was manual and opaque; and newborn development tracking was fragmented across physical records. Additionally, from a business operations perspective, the marketing team had no way to update app content or personalisation settings without requiring a full software release. The project therefore required both a patient-facing product and an internal admin portal to solve these problems holistically.

## 3.2 Challenges Faced During the Internship

### 3.2.1 Challenging Client Requirements

A recurring challenge was managing client expectations around what could realistically be delivered within the agreed timeline and budget. For example, during the Masan display tracking project, the client requested a 98% AI detection accuracy rate under a tight delivery schedule. Achieving such a level of precision with a newly deployed model in a live, uncontrolled retail environment required careful negotiation, setting realistic milestones, and managing the client's expectations transparently throughout the project.

### 3.2.2 Vague and Incomplete Requirements

Clients often provided requirements that lacked the specificity needed for technical implementation. Rather than concrete definitions, initial briefs were frequently high-level and ambiguous. Extracting the precise details required extensive follow-up — for instance, when designing the user interface for the GeneSolutions mobile application, it was often necessary to ask detailed questions about every screen: what elements were to be included, what each button should trigger, what the next page in a flow should look like, and so on. This process of progressive clarification was time-consuming but essential to avoid costly rework during development.

### 3.2.3 Communication Gap with Non-Technical Clients

Many client stakeholders did not have a technical background and were therefore unable to communicate directly with AIVision's engineering team. This created an indirect communication chain in which clients would relay requirements through their own internal

BA contacts, who would then communicate with me as the point of contact at AIVision. Each additional layer introduced the risk of information being lost, distorted, or delayed, making it critical to document every discussion carefully and confirm understanding at each step.

### 3.2.4 Scope Creep and Mid-Project Requirement Changes

A persistent challenge across multiple projects was scope creep — clients requesting new features or changes after the scope had already been formally agreed in the contract. While some requests were reasonable and accommodatable, others were not feasible within the existing timeline or budget. Managing these situations required clear communication about the contractual boundaries and, when appropriate, negotiating a formal scope change process rather than absorbing requests silently.

### 3.2.5 Expectation of Constant Availability

Several clients expected immediate responses during evenings, weekends, and outside of standard working hours. While responsiveness is important in a client-facing role, this created pressure and required setting professional boundaries while maintaining a positive working relationship.

## 3.3 A Specific Challenge faced: AI Detection Failure in the Masan Project

One of the most significant challenges encountered during the internship was a critical AI detection failure that emerged mid-production on the Masan display tracking project.

At the outset, the client had agreed to the operating assumptions of AIVision's detection algorithm, which was designed to work with a single row of SKUs arranged on each shelf of a display rack. Initially, the system performed as expected. However, as deployment scaled, a growing volume of complaints about low detection accuracy was received. Upon investigation, it was discovered that store staff at many locations had been stacking multiple rows of products on a single shelf — a real-world behaviour that fell outside the algorithm's designed operating conditions and caused widespread misdetection.

To resolve the issue, the following steps were taken. First, the problem was diagnosed and documented in detail, including the specific request numbers, timestamps, incorrectly detected SKUs, and recorded accuracy figures. Second, the situation was escalated internally to the development team, with a clear explanation of the root cause and a proposed solution involving a different detection algorithm better suited to variable stacking patterns. Third, a meeting was arranged with the Masan team to transparently communicate the problem, explain the cause, and present the proposed fix. Ultimately, the client was receptive to the solution, and following their agreement, the updated model was deployed. Detection accuracy returned to an acceptable level and complaints subsided.

This experience highlighted the importance of proactive monitoring, clear internal escalation, and transparent client communication when production issues arise.


Figure 3.1: AI mis-detection caused by double-row SKU stacking on a Masan retail display.

As illustrated in Figure 3.1, the Kokomi and Omachi instant noodle packages were stacked in two rows on a single shelf rather than one. Since the detection algorithm was designed under the assumption of a single product row per shelf, products were miscounted and Planogram compliance scores were reported inaccurately. This real-world stacking behaviour, which had not been accounted for in the original deployment agreement, was the root cause of the widespread accuracy complaints received from Masan's field network.

# Chapter 4

# Technologies, Instruments, and Programming Frameworks

This chapter describes the tools, frameworks, and platforms exercised throughout the internship, grouped by their primary purpose.

## 4.1 Wireframing and UI Design

### 4.1.1 Figma and FigJam

Figma is a browser-based, collaborative interface design tool widely used in the software industry for wireframing, prototyping, and high-fidelity UI design. FigJam is Figma's companion whiteboarding tool, designed for brainstorming, flowcharting, and real-time team collaboration. Both tools were used extensively during the GeneSolutions mobile application project. Figma was the primary workspace for constructing wireframes, designing screen mockups, and defining button flows and navigation logic. FigJam was used for mapping out user flows and collaborating with frontend and backend developers to align on interface logic before implementation. Completed designs were also presented to the client directly from Figma during review and demo sessions.

### 4.1.2 Stitch AI

Stitch is an AI-powered UI design tool developed by Google that generates interface designs from natural language prompts. Rather than starting from a blank canvas, a textual description of a screen's purpose and content can be provided, and Stitch produces a first-draft UI layout automatically. During the GeneSolutions project, Stitch was used as a starting point to accelerate the early design phase — prompting it to generate initial screen drafts, which were then imported into Figma for refinement, adjustment to the client's brand guidelines, and further iteration. This workflow significantly reduced the time spent on initial layout construction.

### 4.1.3 React and TypeScript

React is a widely adopted JavaScript library for building component-based user interfaces, and TypeScript is a statically typed superset of JavaScript that improves code reliability and developer tooling. In addition to the Figma-based design workflow, React and TypeScript were used to vibe-code UI screens for the GeneSolutions mobile application. This approach — using an AI-assisted coding workflow to generate interface code from design intent — offered a faster alternative to manual Figma design for certain screens, allowing UI to be prototyped and iterated at a higher pace without being bottlenecked by pixel-level design work.

## 4.2 Project Management

Azure DevOps is a Microsoft platform that provides an integrated suite of tools for software project planning, version control, and delivery management. The primary feature used during the internship was the Boards module, which supports Agile sprint-based project management. This included creating and managing sprint cycles, defining and assigning tasks and subtasks to team members, setting priorities and deadlines, and tracking the status of each work item (e.g. To Do, In Progress, Done) across multiple projects simultaneously. Azure DevOps Boards provided a shared, real-time view of project progress for both the internal team and, where applicable, the client's project contacts.

## 4.3 Requirements Gathering and Documentation

Microsoft Office — primarily Word and Excel — was used for formal documentation throughout the internship. Word was used to produce technical requirements specifications, functional guidelines, and meeting minutes following client and internal team discussions. Excel served as the primary tool for structured data management, covering a wide range of tasks including bug report tracking, SKU tracking sheets, client question lists, logging detected accuracy issues, and maintaining reference lists for quality control purposes.

## 4.4 Quality Control

Label Studio is an open-source data labelling platform used to annotate training and evaluation datasets for AI and machine learning models. During the internship, it was used as part of the monthly quality control process for the Masan AI Display Tracking project. Specifically, Label Studio was used to review and correct incorrect bounding box annotations — cases where the AI model had mis-detected object boundaries or incorrectly identified SKU rows on retail display shelves. The corrected annotations were then compiled into a QC report and submitted to Masan for monthly tracking and reporting. The development team also leveraged this report to support model improvement.

## 4.5 AI Tools

### 4.5.1 NotebookLM

NotebookLM is an AI research and note-taking tool developed by Google that allows users to upload documents and query them conversationally. It was used during requirements gathering to analyse large volumes of client-provided documentation — extracting relevant functional requirements, summarising key points, and identifying gaps or ambiguities. NotebookLM was also used to process audio recordings of client meetings and generate structured meeting minutes, significantly reducing the time required for post-meeting documentation.

### 4.5.2 Claude

Claude is a large language model developed by Anthropic, used during the internship as an AI writing assistant for technical documentation. It was particularly useful for drafting and refining client-facing documents, including functional requirement guidelines, question lists sent to clients during the requirements gathering phase, SKU tracking tables, and bug report summaries. Claude's ability to maintain context across long documents made it well-suited to producing consistent, professional-quality technical writing.

### 4.5.3 ChatGPT and Gemini

ChatGPT (developed by OpenAI) and Gemini (developed by Google) were used as general-purpose AI assistants throughout the internship for day-to-day tasks such as drafting emails, looking up technical concepts, and supporting quick analysis or summarisation tasks. While their usage was less specialised than NotebookLM or Claude, they served as versatile productivity tools that reduced the time spent on routine cognitive tasks.

# Chapter 5

# Tasks Assigned

## 5.1 Wireframing and UI Design

A significant portion of the internship was dedicated to UI/UX design for the GeneSolutions pregnancy mobile application. The project required designing the interfaces for three distinct features: a Baby Naming tool, an After Birth tracking suite, and an internal Admin Portal. All three were delivered through a consistent iterative workflow — translating written requirements into screen designs, gathering client feedback, and refining until a prototype was ready for demo.

### 5.1.1 Design Workflow

The design process for each screen followed a consistent three-stage pipeline. First, the functional requirements for the screen were reviewed and distilled into a structured prompt, which was used either with Google Stitch AI or directly in VS Code using React and TypeScript via an AI-assisted vibe-coding approach to generate a first-draft layout. This initial draft served as a low-effort starting point rather than a blank canvas, allowing rapid exploration of layout options. Second, the generated screens were imported and arranged on a Figma or FigJam board, where navigation flows, button logic, and screen transitions were mapped out and refined. Third, the resulting prototype was reviewed by the client in sessions held approximately two to three times per week. Early review cycles focused primarily on high-level flow and structure, while later rounds narrowed to finer details such as label wording, component spacing, and interaction states. This iterative loop continued for approximately three weeks until a stable prototype was approved for handoff to the development team.

### 5.1.2 Baby Naming Tool

The Baby Naming tool is a personalised name recommendation feature that combines Numerology and Feng Shui to help expecting mothers find a meaningful name for their child. The designed screens covered the full user flow across six stages.

The entry screen allows mothers to select their preferred naming method — Numerology (based on birth date) or Feng Shui (based on birth year and elemental destiny). A detailed input form then collects the baby's gender, the family surname, the mother's preferred naming style, the expected or actual birth date, and a dropdown of blessings or character traits the mother wishes for her child.

Following submission, the app generates a curated list of name suggestions with a brief summary visible inline and a full analytical breakdown accessible on tap. A history log screen allows mothers to revisit previously generated lists. The deep analytics view presents a four-part breakdown of the selected name, covering the linguistic and emotional story behind the name, a Pythagoras birth chart decoding, an integrated name-and-birth-date compatibility grid, and personalised parenting insights derived from the child's numerology profile. Finally, a shareable "Certificate of Love" screen was designed to allow mothers to export and share a visual keepsake of the baby's name and its meaning to social platforms. The complete screen flow is illustrated in Figure 5.1.

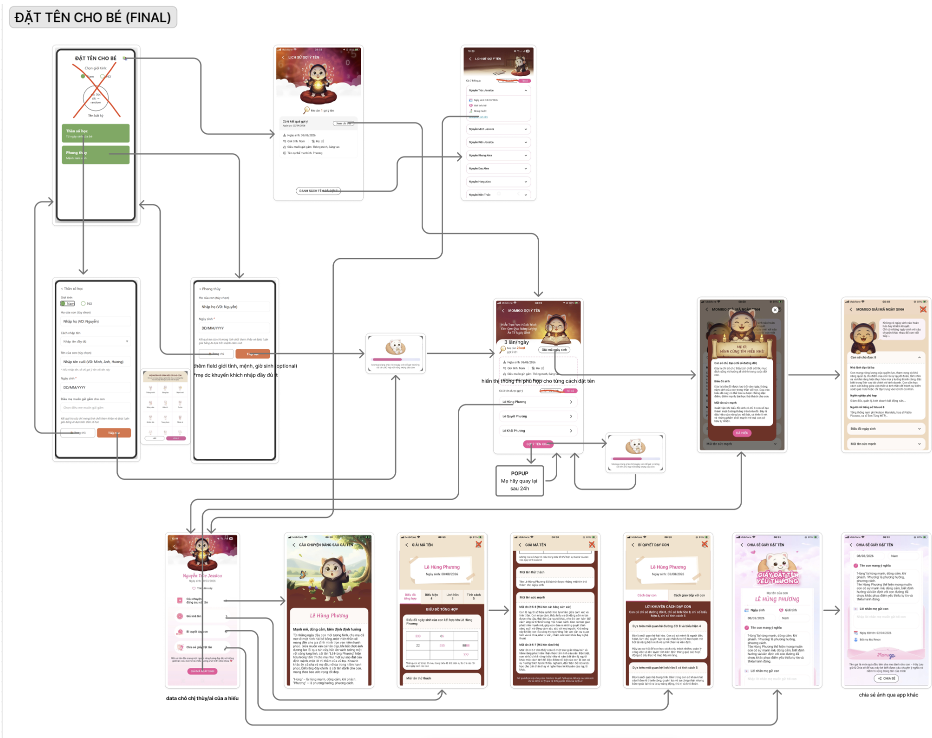


Figure 5.1: FigJam flow diagram for the Baby Naming tool. Click here to view the interactive FigJam board.

### 5.1.3 After Birth Suite

The After Birth suite activates dynamically once a mother records that she has given birth. At 41 weeks of pregnancy, a pop-up prompts the mother to confirm delivery, which triggers a transition from prenatal tracking mode into a postpartum dashboard personalised to the newborn.

The onboarding flow guides the mother through setting up the baby's profile — uploading a photo, entering the full name, gender, birth date and time, and initial weight and height measurements. The central dashboard provides a real-time health summary showing the baby's exact age in months and days, the next upcoming vaccination with a countdown, and the latest growth snapshot with delta indicators from the previous entry. A multi-child management screen was also designed to allow mothers to switch between profiles for multiple children.

The Growth Tracking module features interactive weight and height charts plotted over a monthly timeline, with WHO growth standard benchmarks overlaid to automatically classify the baby's development as within the normal range or flagged as below average. An entry form allows new measurements to be logged by date, with an auto-generated quick analysis summary.

The Immunisation Scheduler organises all required vaccines month by month, explicitly naming each critical shot — including Hepatitis B, the 5-in-1 Quinvaxem/Pentaxim, Rotavirus, Pneumococcal, etc. — to ensure no doses are missed. Finally, the Milestone Journal provides a visual timeline feed where mothers can log dated entries with photos and videos, categorised by milestone type, health events, or sleep records. The full postpartum flow is illustrated in Figure 5.2.

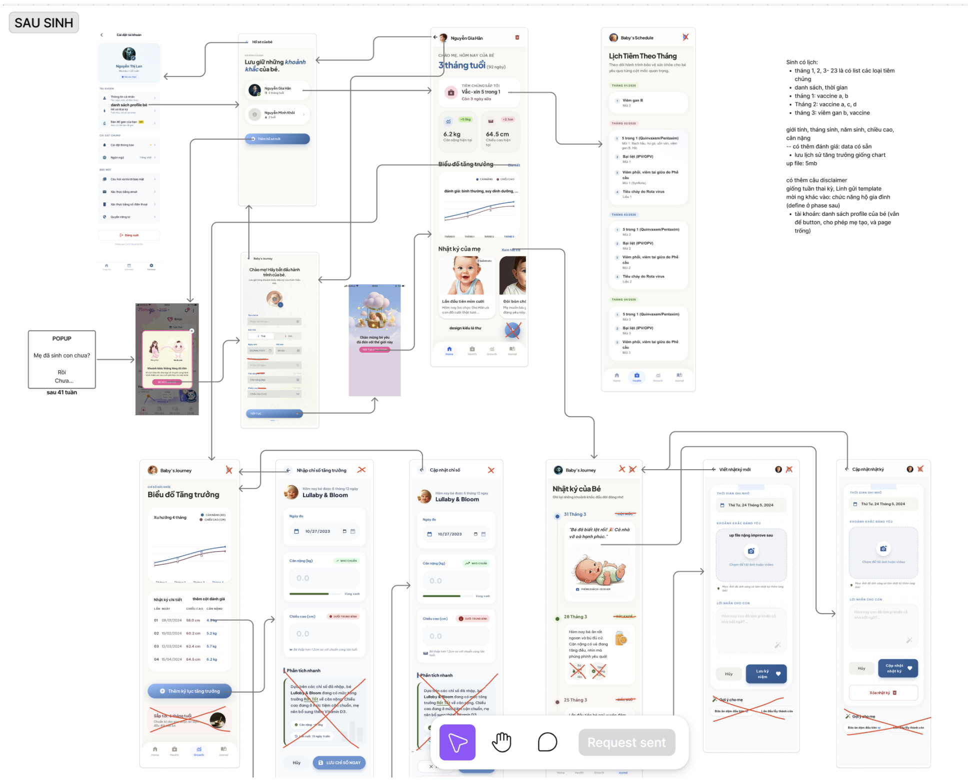


Figure 5.2: FigJam flow diagram for the After Birth suite. Click here to view the interactive FigJam board.

### 5.1.4 Admin Portal

The Admin Portal is a web-based Content Management System (CMS) designed for use by the GeneSolutions operations and marketing team. It provides centralised control over all content, media, clinical surveys, and user accounts within the mobile application — removing the dependency on software releases for routine content updates.

The portal's authentication screen provides a secure login entry point restricted to credentialled GeneSolutions staff. Upon login, a welcome dashboard greets the administrator and defers detailed analytics to a future release phase.

The Handbook Management module supports two content pipelines: native article creation, where admins write and publish articles from scratch using a rich text editor with category and author assignment; and Sale Portal fetching, which pulls pre-vetted corporate content from a linked business platform. Fetched articles have their core fields permanently locked to protect brand and medical accuracy, with admins only permitted to configure app-specific metadata such as categories and target pregnancy stage tags.

The Media Library module covers three sub-sections: banner management with drag-and-drop reordering, a general photo asset repository, and a theming dashboard for applying visual templates across the application.

The Clinical Checklist module provides a form builder interface for creating and versioning clinical surveys, with per-step and per-field configuration including field type, validation rules, and optional or mandatory toggles.

Finally, the User and Admin Access Control module maintains separate registries for admin accounts and end-user accounts, with a detailed per-user view exposing account metadata and synchronised health parameters such as weight, height, blood type, and current pregnancy week. The portal flow is illustrated in Figure 5.3.

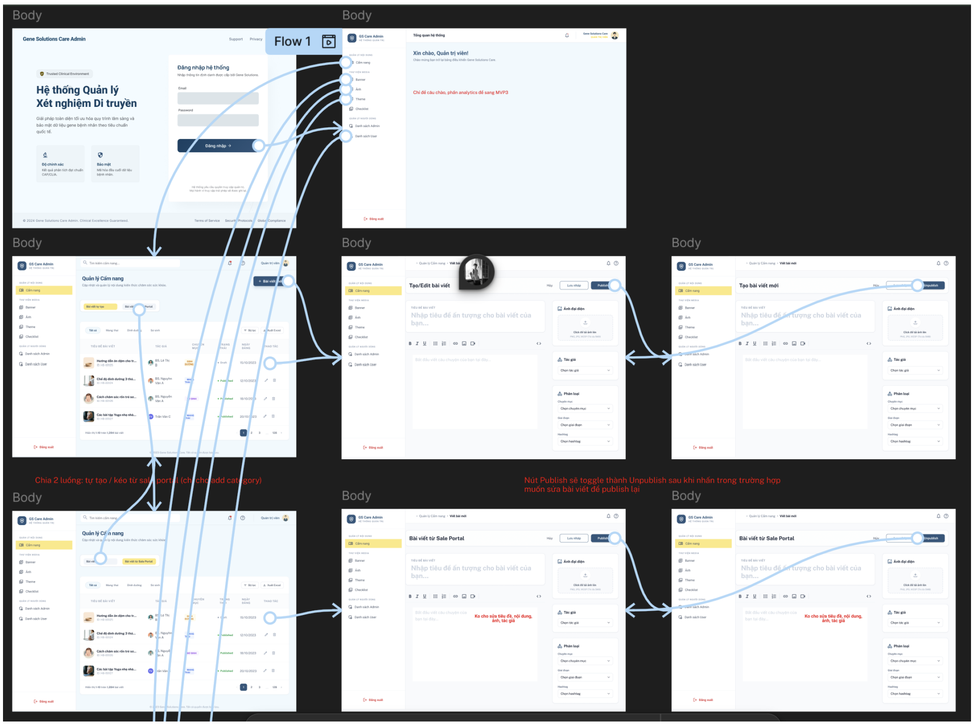


Figure 5.3: FigJam flow diagram for the GS Care Admin Portal. Click here to view the interactive Figma wireframe.

## 5.2 Project Management

## 5.3 Requirements Gathering and Documentation

## 5.4 Quality Control

# Chapter 6

# Intra-Organization Communication

# Chapter 7

# Results Produced and Skills Developed

## 7.1 Results Produced

## 7.2 Skills Developed

# Chapter 8

# Lessons Learned and Personal Comments

## 8.1 Lessons Learned

## 8.2 Personal Comments

# Chapter 9

# Conclusions

# List of References